

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

A) Create a file by using

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill3.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ |
```

Then press enter and the code terminal will open then write a code

```
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main()
{
    char command[100];

    while (1)
    {
        printf("Enter command (type exit to quit): ");
        scanf("%s", command);

        if(strcmp(command, "exit") == 0)
        {
            printf("Exiting program...\n");
            break;
        }

        printf("You entered: %s\n", command);
    }

    return 0;
}
```

After writing the code save the your program by ctrl O and for coming back to command press ctrl X then compile it by the command

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill3.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc skill3.c -o skill3|
```

After compiling to get the output enter command

```

vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill3.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc skill3.c -o skill3
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ./skill3
Enter command (type exit to quit): ls
You entered: ls
Enter command (type exit to quit): pwd
You entered: pwd
Enter command (type exit to quit): exit
Exiting program...
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ |

```

Task 2: Using the following system calls, copy the content from File1.txt to File2.txt.

1. **Open()**
2. **Read()**
3. **Write()**
4. **Close()**

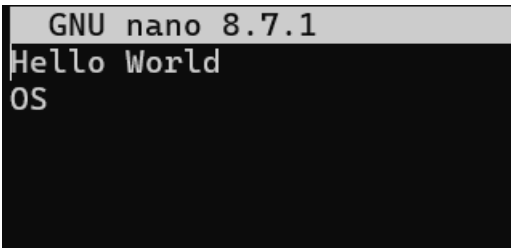
Flag	Purpose
O_RDONLY	Open for read only
O_WRONLY	Open for write only
O_RDWR	Open for both read and write
O_CREAT	Create file if it does not exist
O_TRUNC	Truncate (empty) an existing file
O_APPEND	Append data to the end of the file
O_EXCL	Fail if file already exists (used with O_CREAT)

1) Create a file

```

vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill3main2.txt
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ |

```



2) Create another file

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill3main2.txt
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill3main2.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ |
```

```
GNU nano 8.7.1 skill3main2.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int source, destination;
    char buffer[100];
    int bytes;

    source = open("file1.txt", O_RDONLY);

    destination = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);

    while((bytes = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, bytes);
    }

    close(source);
    close(destination);

    printf("File copied successfully.\n");

    return 0;
}
```

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill3main2.txt
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill3main2.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc skill3main2.c -o skill3main2
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ./skill3main2
```

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ cat skill3main2.txt
Hello World
OS
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ |
```